

Model Development & Governance Document

Model Name: Credit Risk Scorecard (PD) Model v1.0

Owner: Quantitative Risk Control Group

Effective Date: June 2026

1. Objective & Scope

The objective of this scorecard is to predict the Probability of Default (PD) for retail credit loan applications. The model is scaled to scorecards suitable for automated real-time credit decisions. The scope covers all unsecured, individual loans.

2. Model Development Methodology

The model uses a Logistic Regression framework fitted on Weight of Evidence (WoE) binned variables. Predictors were selected using Information Value (IV) thresholds between 0.02 and 0.50. This linear formulation is highly interpretable, audit-compliant under Basel framework, and robust to outliers.

3. Model Variables & Selected Features

Selected Variable	Variable Type	IV Value	Role in Model
annual_inc	numeric	0.0427	Predictor Feature
revol_util	numeric	0.0818	Predictor Feature
inq_last_6mths	numeric	0.0579	Predictor Feature
pub_rec	numeric	0.0300	Predictor Feature
pub_rec_bankruptcies	numeric	0.0226	Predictor Feature
purpose	categorical	0.0610	Predictor Feature

4. Model Validation Performance

Model validation was performed using In-Time test data and an Out-of-Time (2011) test sample. The model's Gini coefficient is **0.262**, the KS statistic is **0.187**, and the Brier score is **0.1308**. The OOT validation confirmed temporal stability with a total score PSI of **0.0091**.

5. Reject Inference & Sample Selection Bias

A recognized limitation of this model is sample selection bias. The training population consists only of approved loans, leaving the performance of rejected applicants unobserved. Re-weighting or parceling reject inference techniques will be investigated in subsequent reviews.

6. Model Governance and Monitoring Plan

The model will be monitored monthly for score stability (PSI) and quarterly for performance degradation (Gini/KS drops). If the score PSI exceeds 0.25, or if the KS statistic falls below 0.30, an automatic model retraining/redevelopment trigger will execute. Model audits will be performed annually.